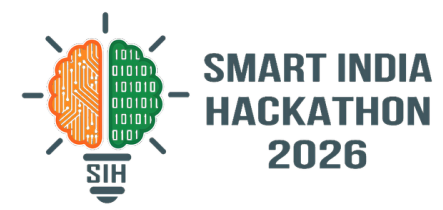
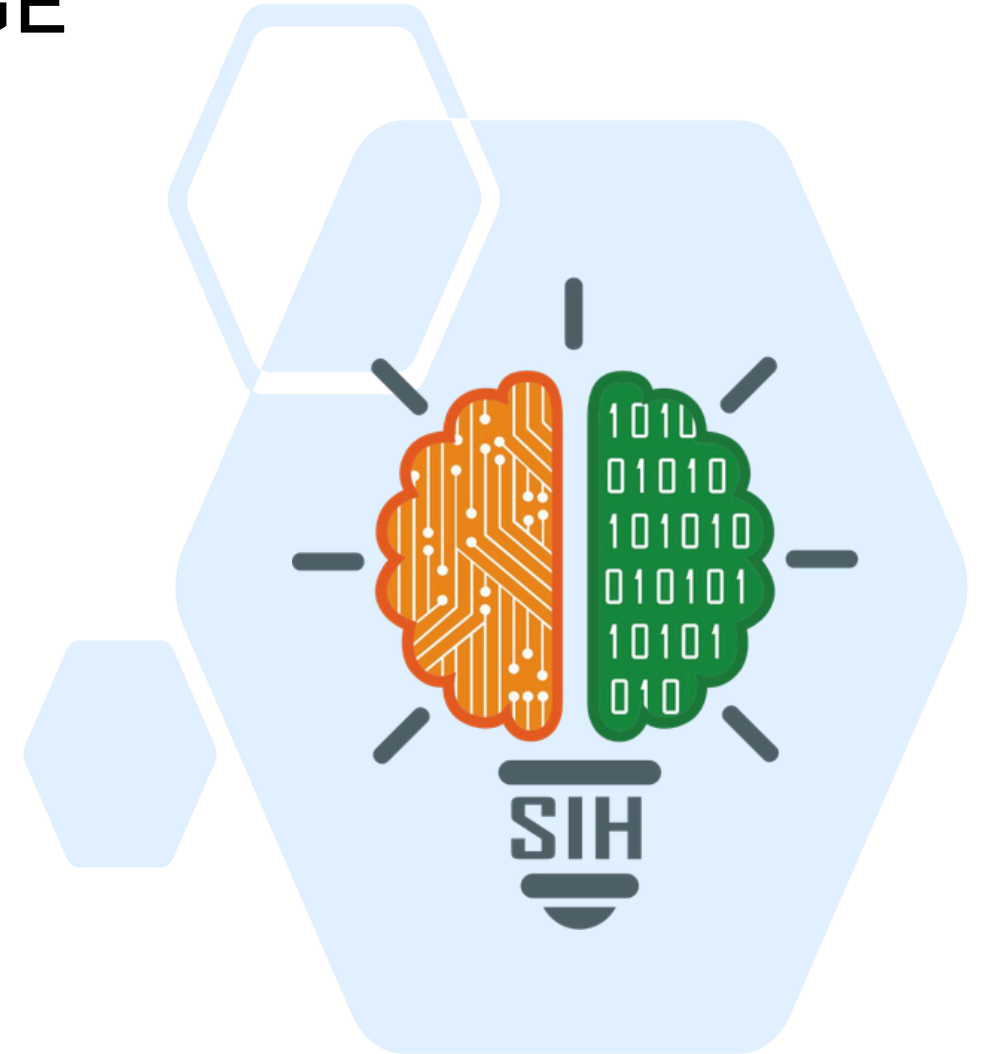


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26169
- **Problem Statement Title** – AI-Based Virtual Camera Tracking
System for Coarse Alignment of Mobile FSOC Terminals
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – —
- **Team Name** – ZeroDrift





IDEA TITLE

ZeroDrift

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

The eyes and neck of a laser-communication terminal

ISRO · PS 26169
every figure measured

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

1 SEE

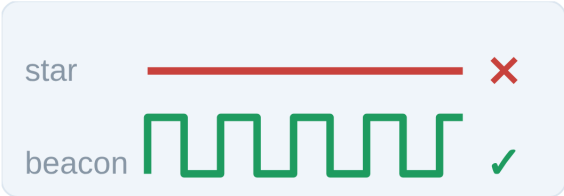
A physics-true sky: stars, clutter, glint, turbulence.



30 fps · real ISS orbit

2 IDENTIFY

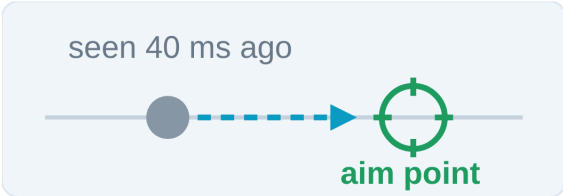
The beacon blinks at 4 Hz. A star cannot.



0 wrong locks in 64 runs

3 PREDICT

Aim where it will be, not where it was.



40 ms of delay cancelled

4 STEER

Correct every frame — 30 times a second.



8–21 ms of the 33 ms budget

INNOVATION AND UNIQUENESS OF THE SOLUTION

Brightness is not identity — modulation is.

Star / glint
brighter, steady



REJECTED

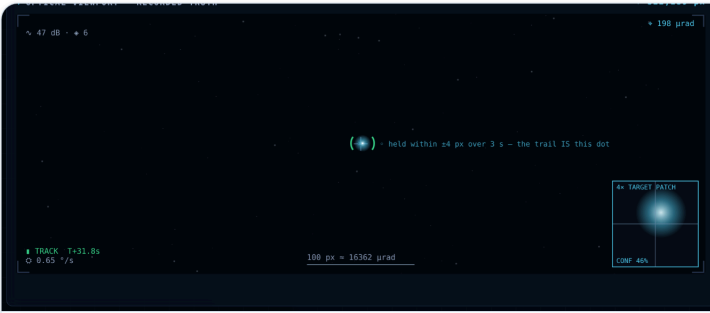
Our beacon
dimmer, 4 Hz



LOCKED

A decoy can be brighter — it still cannot blink at 4 Hz.

HOW IT ADDRESSES THE PROBLEM



Our live console, mid-pass on a real ISS orbit.

1.27s
TO ACQUIRE

98.3%
LOCK HELD

0/64
WRONG LOCKS

TECHNICAL APPROACH

CORE



Python 3.11 · NumPy · OpenCV

INTERFACE



PySide6 console · pyqtgraph live plots

ORBITS · AI



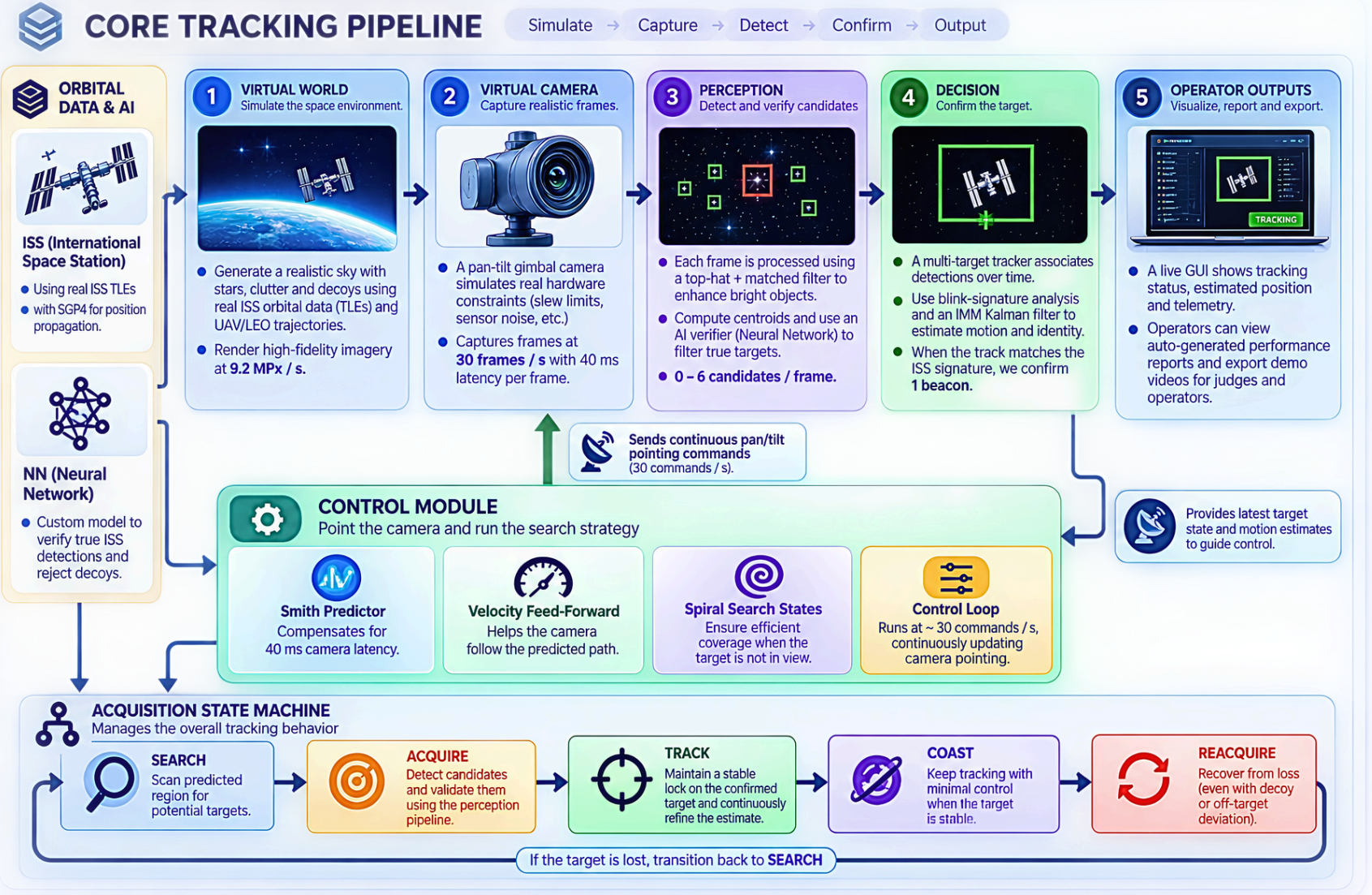
real ISS TLEs · Neural Network

QUALITY



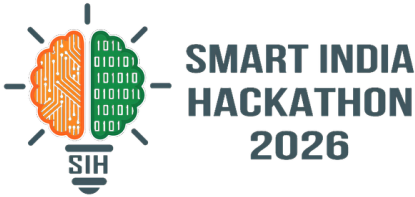
pytest suite · GitHub Actions

SHIPPING





FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

not a plan — a working system

100%

73 / 73

automated tests pass

100%

64 / 64

runs acquired the beacon

✓

0 wrong locks

in 64 randomised runs

8–21 ms

of the 33 ms frame

headroom

no GPU · no cloud · no dataset

POTENTIAL CHALLENGES AND RISKS

- Heavy turbulence

dims and shakes the beacon

→
- Decoys & sun-glint

brighter than the real beacon

→
- 40 ms mount latency

loops oscillate and lag

→
- Sharp UAV manoeuvres

simple filters fall behind

→

STRATEGIES FOR OVERCOMING THESE CHALLENGES

- Coast through the fade, then re-spiral

1.27 s · 94.3 % lock in hard turbulence
- 4 Hz blink gate + AI verifier

0 / 64 wrong locks
AUC 0.957 vs 0.900
- Two-path Smith predictor

4.2 μrad lag at 0.75 °/s
- Dual-model IMM estimator

re-weighted every frame

Worst of 64 runs not the average	4.0 s slowest acquisition median 1.27 s	91.6 % lowest lock held median 97.6 %	2.8 s dim beacon, heavy turbulence — still acquired	1 command regenerates every figure
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IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

two-stage architecture, justified by measurement

Does the optical link actually close?

Fraction of a real ISS pass with the modelled link closed.



6.8× more link time

Tracked ISS pass · 250 μrad beam · 3.1 dB margin · zero outages.
Measured, not asserted.

What it replaces

An optical test bench

₹ lakhs

Cameras, gimbals, optics, lab time. One team at a time.

ZeroDrift

₹ 0

A laptop you already own. One command. A whole class at once.

- no GPU
- no cloud
- no dataset
- no consumables
- open source

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

STRATEGIC

Sovereign FSOC pointing capability, developed in India.

ECONOMIC

Lakhs of rupees of hardware → zero recurring cost.

RESEARCH & EDUCATION

Any lab or student can test PAT algorithms today.

SCALABLE & GREEN

Satellite, UAV, ground, maritime — one pure-software engine.

Who it serves:

ISRO / IN-SPACe optical-communication teams

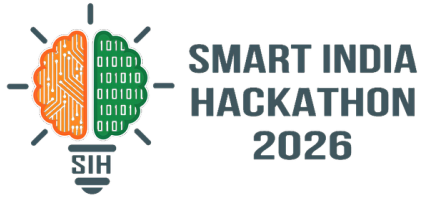
DRDO

IITs & engineering colleges

India's space-tech start-ups



RESEARCH AND REFERENCES



Research foundations

Kaymak et al., “A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,” IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)

Kaushal & Kaddoum, “Optical Communication in Space: Challenges and Mitigation Techniques,” IEEE Comm. Surveys & Tutorials, 2017. [\[link\]](#)

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management. [\[link\]](#)

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)

Smith, O.J.M., “Closer Control of Loops with Dead Time” — the Smith predictor, 1957. [\[link\]](#)

Vallado & Crawford, “SGP4 Orbit Determination” — AIAA 2008; real ISS TLE propagation. [\[link\]](#)

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation models. [\[link\]](#)



Our work —open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Live demo (replay console): zerodrift-fsoc-pat.netlify.app

Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo campaign logs · technical report · user manual · demo video.

Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods

Compared with existing approaches			
Feature	ZeroDrift	MATLAB sim	Traditional
Zero hardware cost No special equipment required	✓	✓	✗
Real-time closed loop + live console See it run. Live.	✓	✗	✓
Decoy & blink-signature discrimination Robust against spoofing	✓	✗	✗
Open source • one-command reproducible Run with a single command	✓	✗	✗